Krishna Koushik Maddukuri

Senior Full Stack Software Engineer | Java · Spring Boot · Apache Kafka · AWS · Angular · React

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PROFESSIONAL SUMMARY

Senior Full Stack Software Engineer with 6+ years of experience designing, building, and operating large-scale distributed systems and cloud-native microservices for financial services firms including Bank of America and Capital One. Expert in Java, Spring Boot, Apache Kafka, REST API design, system design, Angular/React, and Amazon Web Services (AWS), with a record of delivering secure, regulatory-compliant, high-throughput data platforms. Reduced batch processing time by 43%, cut deployment time by 70%, and lowered compute costs through container optimization. AWS Certified Developer – Associate with deep experience in event-driven architecture, CI/CD automation, test-driven development (TDD), and production observability.

TECHNICAL SKILLS

Languages: Java (8/11/17), Python, SQL, TypeScript, JavaScript, C++, Bash, Groovy

Backend & Frameworks: Spring Boot, Spring MVC, Spring Batch, Spring Kafka, Spring Security, Spring Data JPA, Hibernate, Microservices, Distributed Systems, System Design, RESTful APIs, OpenAPI/Swagger, GraphQL, OAuth2, JWT, Node.js, Liquibase

Cloud & Infrastructure: AWS (ECS, Fargate, EC2, Lambda, S3, SNS, SQS, RDS, DynamoDB, Step Functions, CloudFormation, CloudWatch, IAM, Glue, EMR), Microsoft Azure, Docker, Kubernetes, Serverless Architecture, Infrastructure as Code (IaC)

Data & Messaging: Apache Kafka, PostgreSQL, MySQL, SQL Server, DynamoDB, MongoDB, Event-Driven Architecture, Real-Time Data Pipelines, ETL, Metro 2 Credit Reporting

DevOps & CI/CD: Jenkins, Git, Bitbucket, GitHub, Maven, Gradle, SonarQube, Blue-Green Deployments, Zero-Downtime Releases, Agile/Scrum, Jira, Confluence

Testing & Quality: JUnit, Mockito, Cucumber, Selenium, WireMock, Postman, Test-Driven Development (TDD), Behavior-Driven Development (BDD), Code Review

Observability & Security: Splunk, New Relic APM, Prometheus, ELK Stack, Spring Actuator, Log4j, Performance Tuning, Secure SDLC, Regulatory Compliance

Frontend: Angular, React.js, Redux, TypeScript, HTML5, CSS3, Bootstrap, jQuery, Jasmine

PROFESSIONAL EXPERIENCE

Application Programmer III (Full Stack Software Engineer) **Oct 2026 – Present**

Bank of America *Pennington, NJ*

- Develop and maintain full-stack features for Bank of America's Wealth Management Technology platform: Java/Spring Boot microservices and RESTful APIs on the back end, Angular and React front ends, delivering secure, scalable, high-availability financial applications.
- Build event-driven integrations with Apache Kafka and relational data stores, following the bank's secure SDLC, code review, and regulatory compliance standards.
- Write automated unit and integration tests (JUnit, Mockito, Jasmine) and ship through CI/CD pipelines, collaborating in Agile/Scrum ceremonies with product, QA, and platform teams.

Senior Software Engineer **Sep 2024 – Oct 2026**

Capital One *McLean, VA*

- Designed and built scalable, secure Java/Spring Boot microservices for Capital One's Enterprise Credit Bureau Reporting Service, generating compliant Metro 2 files for approximately 30 million consumer, small-business, and commercial accounts each month across Equifax, Experian, TransUnion, and LexisNexis.
- Engineered high-throughput, event-driven pipelines with Apache Kafka, Spring Kafka, and PostgreSQL on Amazon RDS, achieving fault-tolerant, exactly-once processing of sensitive financial data at enterprise scale.
- Improved batch performance by 43% (7 hours to 4) through performance tuning of Spring Batch readers, replacing cursor-based readers with a paginated reader for high-volume Metro 2 file generation.
- Deployed containerized services on AWS Fargate via CloudFormation, integrating SNS, S3, RDS, and Step Functions; reduced container image size by 60%, lowering compute costs and improving runtime efficiency.

- Implemented enterprise CI/CD pipelines in Jenkins to automate builds, enforce SonarQube quality gates, and run blue-green deployments with zero-downtime releases.
- Strengthened security and compliance by designing OAuth2-protected REST APIs and building automated test suites with JUnit and Cucumber, reducing production defects.
- Integrated Splunk and New Relic APM with actionable dashboards and alerting, accelerating incident response and root-cause analysis across distributed microservices.
- Onboarded Discover account portfolios into the credit bureau reporting pipeline following the Capital One–Discover merger, extending the platform to new product segments without disrupting monthly regulatory deadlines.

Software Engineer **Sep 2023 – Sep 2024**
NGA Group Inc *New Brunswick, NJ*

- Designed a cloud-ready Java/Spring Boot service unifying disparate data sources into a standardized reporting model, improving data accessibility and eliminating manual reconciliation across analytics workflows.
- Delivered secure, role-based REST APIs enabling downstream teams to consume curated datasets without direct database access, strengthening data governance.
- Built a high-velocity ingestion pipeline with Apache Kafka and DynamoDB, supporting near-real-time propagation of updates and scaling with increased upstream event volume.
- Containerized the application on AWS ECS with Docker and automated build and release pipelines in Jenkins, reducing deployment time by 70% and improving release reliability.
- Improved operational visibility by integrating Spring Actuator, Prometheus, and ELK analytics, cutting issue-detection time by 50%.

Software Engineer **Aug 2020 – Aug 2021**
Infosys Limited *Hyderabad, India*

- Developed modular, reusable React.js components with Redux state management, delivering consistent UI behavior across 4 large-scale enterprise applications for a global retail client.
- Integrated RESTful APIs with React and Redux for asynchronous data fetching and built front-end features (forms, routing, events, custom components) backed by Java/Spring MVC services.
- Modernized legacy codebases and applied engineering best practices, reducing bug rates by 25% and improving maintainability and performance.
- Implemented CI/CD pipelines with Bitbucket and automated multi-region AWS deployments, increasing deployment frequency by 3x.

Software Engineer Intern **Jan 2020 – Aug 2020**
Infosys Limited *Mysore, India*

- Completed 15+ projects and certification-aligned exercises across back-end and front-end development within the MERN ecosystem (MongoDB, Express, React, Node.js).
- Diagnosed and resolved backend build failures by analyzing Jenkins logs and debugging deployment scripts in Bash and Groovy, improving CI/CD pipeline stability by 40%.
- Coordinated sprint planning for a four-developer team using Agile methodology and Jira, ensuring on-time delivery of project milestones.

EDUCATION

Doctor of Business Administration (DBA), Technology Management Jul 2026 – Aug 2029 (expected)
Westcliff University

Master of Science (M.S.), Computer Science 2021 – 2023
George Mason University, Fairfax, VA

Bachelor of Technology (B.Tech), Computer Science 2016 – 2020
Amrita Vishwa Vidyapeetham, Coimbatore, India

CERTIFICATIONS

- AWS Certified Developer – Associate, Amazon Web Services (issued Nov 2023)
- AWS Certified Cloud Practitioner, Amazon Web Services (issued Jun 2023)